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Executive Summary

In this problem, we take our previous PFD for cyclohexane production where we produced 300 KMTA using the addition of different types of equipment such as compressors, pumps, shell and tube heat exchangers, and drums. Now, we try increasing the cyclohexane recovery by replacing the low pressure separator with a distillation column. Next, we take the list of all utilities and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Furthermore, now that we have a new equipment we have to recalculate our total capital cost for all equipment. Doing this, we attempt maximizing the recovery of cyclohexane and see if it is more cost efficient than having our original PFD with the low pressure separator.

Systems and Procedures

In this problem we take our previous PFD for cyclohexane production where we produced 300 KMTA using the addition of different types of equipment such as compressors, pumps, heat exchangers, and separators. Using the same design specification from the previous PFD, all we changed was removing our low pressure separator with a distillation column.. However, in order to do this we have to a short cut distillation model, which is followed by the rigorous distillation model that will help us maximize the cyclohexane recovery. Before we begin the shortcut model, we used the Sep2 option in Aspen. Using this allows us to specify the pressure at which we want our distillation column to run at. In the Aspen we make sure to input our separator fractions for our components in the distillate and bottoms. In the short cut model we used the regular DISTWU option in aspen and specified our pressure of our condenser and reboiler around the pressure obtained using the Sep2 option. We also inserted our desired reflux ratio is a standard value of 1.15 which is 1.15 times the minimum. Lastly, we inserted our desired Cyclohexane recovery in the high key section and our desired light key recovery. After running this with our shortcut model, we began our rigorous model. Our rigorous model was used with the Radfrac distillation column option in aspen. To specify the operating parameters of the RadFrac column, we copied the number of columns, feed stage, reflux ratio, and the distillate to feed ratio from the results of the short-cut distillation column when we ran it. After this, we added two design specs regarding the distillate to feed ratio and the reflux ratio. We manipulated these handles in order to specify the desired recovery for the cyclohexane. After this, our simulation ran giving us necessary recovery or our maximum recovery. Now that our process had all necessary utilities and equipment, equations were used for each type of equipment we used. Looking at our equations obtained from the Maxwell and Swift index used in appendix 1, you see the different types of equations used for

purchase and installation cost for the shell and tube heat exchangers, pumps, compressors, and drums. Using the installation and purchase cost for every different type of equipment, we obtained the annualized cost for that specific equipment. Addition of every equipment's annualized cost, we see our total annualized cost for the entire cyclohexane production process of approximately ??????. This is compared to the cost of the cyclohexane production when he had a low pressure separator instead of a distillation column. We also compare them to see which gave us the maximum cyclohexane recovery.

Results

1. Problem 5A – Cyclohexane Production

- i. With the given data, we were able to design another aspen simulation where the low pressure separator was replaced with a distillation column,. Doing this we obtained a maximum cyclohexane recovery of ??? and we calculated our new total annualized cost for each equipment including the distillation column inserted . You can also see our total annualized cost for the entire process is about ?????.
- ii. Additional Data:
 - i. PFD – See *Figure 2* in Appendix 2
 - ii. Material Stream Data – See *Table 2* in Appendix 3
 - iii. Heat Data – See *Table 3* in Appendix 3
 - iv. Capital Cost – See *Table 4* in Appendix 3
 - v. Operating Cost- See *Table 5* in Appendix 3

Conclusion

In conclusion for this problem we got to replace the low pressure separator with a distillation column where a shortcut and rigorous model were applied in order to obtain the maximum cyclohexane recovery. Lastly we identified the actual equipment and utilities needed for the cyclohexane production to produce the necessary maximum recovery. Once we had our entire PFD, we calculated our total capital cost using all our equipment with the addition of the distillation column and we found the cost of all utilities in our process and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Doing this by using the equations obtained from the Maxwell and Swift index, we understand how a distillation column replacing the low pressure separator can affect the maximization of the cyclohexane recovery and how it can affect the total annual cost for the entire cyclohexane production.

Appendix 1

Equations and Formulas

Shell and Tube Heat Exchangers:

$$\text{Purchase Cost(2010)} = 28,000 + 54 * A^{1.2}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: } \text{index(2010)} = 1500$$

$$\text{To obtain 'A': } Q = F_t * A * U * \Delta T_{ln}$$

$$\text{Where: } Q = \text{Obtained from Aspen Simulation}$$

$$F_t = 0.8$$

$$U = 50 \frac{\text{Btu}}{\text{hr} * \text{ft}^2 * ^\circ\text{F}}$$

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln (\Delta T_1 / \Delta T_2)}$$

$$\text{Installation Factor} = 3.5$$

$$\text{Installation Cost} = 3.5 * \text{Purchase Cost(2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Pumps:

$$\text{Purchase Cost(2010)} = 8,000 + 240 * Q^{0.9}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: } \text{index(2010)} = 1500$$

$$Q = \text{Obtained from Aspen Simulation in } \frac{L}{s}$$

$$\text{Installation Factor} = 4$$

$$\text{Installation Cost} = 4 * \text{Purchase Cost(2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Compressors:

$$\text{Purchase Cost(2010)} = 500,000 + 20,000 * W^{0.6}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: index(2010)} = 1500$$

W= Obtained from Aspen Simulation in Kw.

$$\text{Installation Factor} = 2.5$$

$$\text{Installation Cost} = 2.5 * \text{Purchase Cost(2020)}$$

$$\text{Annualized Cost} = \frac{4}{3} * \text{Installation Cost}$$

Drums:

$$\text{Assumptions: } \frac{L}{D} = 2, \text{ Residence Time} = 5 \text{ minutes}$$

$$\text{Purchase Cost(2010)} = 11,600 + 34 * M^{0.85}$$

$$\text{Purchase Cost(2020)} = \text{Purchase Cost(2010)} + \frac{\text{index(2020)}}{\text{index(2010)}}$$

$$\text{Where: index(2010)} = 1500$$

$$\text{Where: } M = (\pi * D * L) * \text{Thickness} * \rho_{steel}$$

$$\rho_{steel} = 8000 \frac{kg}{m^3}$$

$$\text{Thickness} = \frac{P_i D_i}{2 * S * E - 1.2 * P_i}$$

$$\text{Where: } E = \text{Welded Efficiency} = 1$$

$$S_{steel} = \text{Stress} = 12.9 \text{ Ksi}$$

$$P_i = \text{Internal Operating Pressure From Aspen Simulation plus 10\% safety factor}$$

$$D_i = \text{Internal Diameter}$$

$$\text{Where: } D_i = \frac{2 * (V_{vessel})^{1/3}}{\pi}$$

Installation Factor= 4

Installation Cost= 4* Purchase Cost(2020)

Annualized Cost= $\frac{4}{3}$ * Installation Cost

Distillation Columns:

-Purchase Cost Trays= 130 + 440(*Diam*^{1.8})

-Purchase Cost Condenser (2010)= 28,000 + 54*A^{1.2}

Purchase Cost(2020)= Purchase Cost(2010) + $\frac{\text{index}(2020)}{\text{index}(2010)}$

Where: *index*(2010)= 1500

To obtain 'A': $Q = F_t * A * U * \Delta T_{ln}$

Where: Q= Obtained from Aspen Simulation

$$F_t = 0.8$$

$$U = 50 \frac{Btu}{hr * ft. * ^\circ F}$$

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln (\Delta T_1 / \Delta T_2)}$$

Installation Factor= 3.5

Installation Cost= 3.5* Purchase Cost(2020)

Annualized Cost= $\frac{4}{3}$ * Installation Cost

-Purchase Cost Reboiler(Kettle)= 29000 + 400(*Area*)^{0.9}

-Reflux Pump= 8000 + 240(*Q*)^{0.9}

-Reflux Drum(Vertical)= Purchase Cost(2010)= 11,600 + 34*M^{0.85}

Purchase Cost(2020)= Purchase Cost(2010) + $\frac{\text{index}(2020)}{\text{index}(2010)}$

Where: $index(2010) = 1500$

Where: $M = (\pi * D * L) * Thickness * \rho_{steel}$

$$\rho_{steel} = 8000 \frac{kg}{m^3}$$

$$Thickness = \frac{P_i D_i}{2 * S * E - 1.2 * P_i}$$

Where: E= Welded Efficiency= 1

$$S_{steel} = \text{Stress} = 12.9 \text{ Ksi}$$

P_i = Internal Operating Pressure From Aspen Simulation plus 10% safety factor

D_i = Internal Diameter

$$\text{Where: } D_i = \frac{2 * (V_{vessel})^{1/3}}{\pi}$$

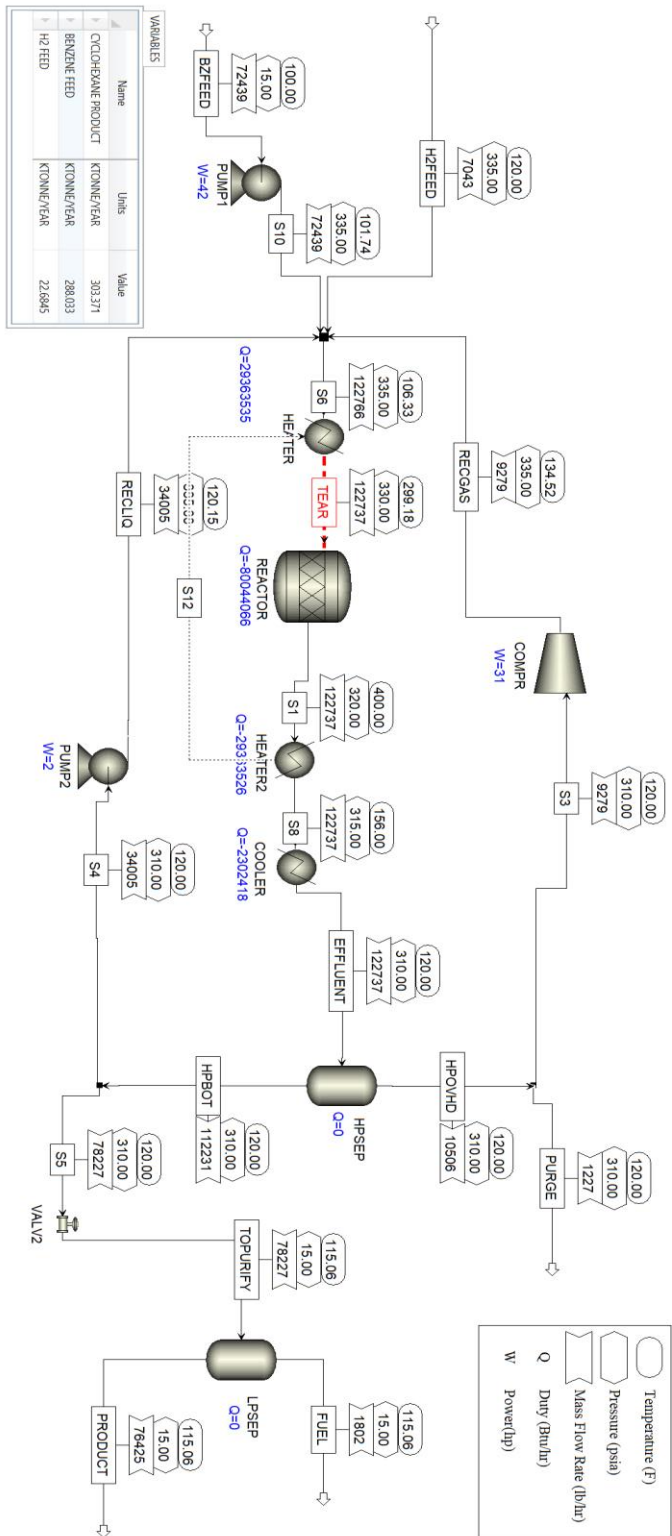
Installation Factor= 4

Installation Cost= 4* Purchase Cost(2020)

Annualized Cost= $\frac{4}{3}$ * Installation Cost

Appendix 2

Figure 1 – Cyclohexane Production PFD



Appendix 3

Energy/Material

Balance

Material													
Stream Name	Units	BZFEED	EFFLUENT	FUEL	H2FEED	HPBOT	HPOVHD	PRODUCT	PURGE	RECGAS	RECLIQ	TEAR	TOPURIFY
From			COOLER	LPSEP		HPSEP	HPSEP	LPSEP	MIX2	COMPR	PUMP2	HEATER	VALV2
To		PUMP1	HPSEP		MIX1	MIX3	MIX2			MIX1	MIX1	REACTOR	LPSEP
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Temperature	F	100	120	115.061	120	120	120	115.061	120	134.521	120.152	299.176	115.061
Pressure	psia	15	310	15	335	310	310	15	310	335	335	330	15
Molar Density	lbmol/cuft	0.686671	0.122001	0.00244952	0.0531551	0.566225	0.0499393	0.558444	0.0499393	0.0525874	0.566164	0.041104	0.0437681
Average MW		78.1136	57.7678	36.2546	2.42641	81.5425	14.0396	84.0169	14.0396	14.0396	81.5425	25.0179	81.5425
Mole Flows	lbmol/hr	927.36	2124.67	49.7002	2902.64	1376.35	748.313	909.636	87.3722	660.941	417.017	4905.98	959.337
H2	lbmol/hr	0	266.194	2.8964	2830.07	4.18636	262.008	0.0215424	30.5917	231.416	1.26841	3061.43	2.91795
N2	lbmol/hr	0	101.888	2.96944	14.5132	4.35305	97.5346	0.0646915	11.388	86.1466	1.31892	101.888	3.03414
C1	lbmol/hr	0	418.076	29.011	58.0527	44.1142	373.962	1.73719	43.6634	330.299	13.366	401.372	30.7482
BZ	lbmol/hr	927.36	1.85607	0.0219139	0	1.83445	0.0216134	1.25672	0.00252356	0.0190899	0.555815	928.034	1.27864
CH	lbmol/hr	0	1336.65	14.8014	0	1321.87	14.7874	906.556	1.72655	13.0608	400.508	413.259	921.358
Mass Flows	lb/hr	72439.4	122737	1801.86	7042.97	112231	10506.1	76424.8	1226.67	9279.38	34004.6	122737	78226.7
H2	lb/hr	0	536.615	5.8388	5705.08	8.43919	528.176	0.0434269	61.6692	466.507	2.55696	6171.48	5.88223
N2	lb/hr	0	2854.23	83.1845	406.565	121.944	2732.28	1.81223	319.018	2413.27	36.9475	2854.23	84.9967
C1	lb/hr	0	6707.1	465.416	931.326	707.713	5999.38	27.8693	700.481	5298.9	214.428	6439.11	493.286
BZ	lb/hr	72439.4	144.984	1.71177	0	143.296	1.6883	98.1673	0.197124	1.49118	43.4168	72492.1	99.8791
CH	lb/hr	0	112494	1245.71	0	111250	1244.52	76296.9	145.309	1099.21	33707.2	34780.4	77542.7
Mass Fractions													
H2		0	0.00437206	0.00324043	0.810039	7.52E-05	0.0502735	5.68E-07	0.0502735	0.0502735	7.52E-05	0.050282	7.52E-05
N2		0	0.0232548	0.0461659	0.0577263	0.00108654	0.260068	2.37E-05	0.260068	0.260068	0.00108654	0.0232548	0.00108654
C1		0	0.0546459	0.258298	0.132235	0.00630585	0.57104	0.000364663	0.57104	0.57104	0.00630585	0.0524625	0.00630585
BZ		1	0.00118126	0.000950003	0	0.00127679	0.000160698	0.0012845	0.000160698	0.000160698	0.00127679	0.590628	0.00127679
CH		0	0.916546	0.691346	0	0.991256	0.118458	0.998327	0.118458	0.118458	0.991256	0.283373	0.991256
Volume Flow	cuft/hr	1350.51	17415.2	20289.7	54606.9	2430.75	14984.5	1628.88	1749.57	12568.4	736.566	119355	21918.6
Molar Enthalpy	Btu/lbmol	21902	-47349	-34060.9	-334.286	-63937.2	-16838.8	-65569.6	-16838.8	-16718.9	-63926.3	2264.36	-63937.2
Mass Enthalpy	Btu/lb	280.386	-819.644	-939.493	-137.77	-784.097	-1199.38	-780.433	-1199.38	-1190.83	-783.963	90.5097	-784.097

Table 1 – Cyclohexane Production Material Balance

Heat	
Stream Name	S12
QCALC Btu/hr	29363535
TBEGIN F	400
TEND F	156

Table 2 – Cyclohexane Production Heat

Cyclohexane Production Cost Estimate						
EQUIPMENT LIST: CAPITAL COSTS						
	Project:	Assignment #4	Chemical Engineers:	Khalil Shehaded	Esteban Vazquez-Elizondo	
	Location:	Chemical Engineering Department	Date:	10/22/2019		
Equipment Class	ASPEN BlockID	Service Description	Tag	Material Cost	Installed Cost	Annual Cost, \$/yr
Distillation						
Drums						
Vertical Tank	HPSEP	High Pressure Separator Drum	V-1	\$340,443.00	\$1,361,772.00	\$1,815,696.00
Vertical Tank	LPSEP	Low Pressure Separator Drum	V-2	\$60,763.70	\$243,054.80	\$324,073.07
Heat Exchangers						
S&T Exchanger	HEATER	Feed-Recycle Exchanger	E-1	\$310,735.00	\$1,087,572.50	\$1,450,096.67
S&T Exchanger	HEATER2	Effluent Cooler	E-2	\$39,957.50	\$139,851.25	\$186,468.33
S&T Exchanger	COOLER	Effluent Cooler 2	E-3	\$42,816.50	\$149,857.75	\$199,810.33
Air Coolers						
Pumps						
Pump	PUMP1	Benzene Pump	P-1	\$8,157.32	\$32,629.28	\$43,505.71
Pump	PUMP2	Recycle Pump	P-2	\$13,324.80	\$53,299.20	\$71,065.60
Turbines						
Compressors						
Compressor	COMPR	Recycle Gas Compressor	C-1	\$632,475.00	\$1,581,187.50	\$2,108,250.00
Reactors						

Table 3 – Cyclohexane Production Cost Estimate

Cyclohexane Production Cost Estimate						
UTILITIES: OPERATING COST						
	Project:	Assignment #4	Chemical Engineers:	Khalil Shehaded	Esteban Vazquez-Elizondo	
	Location:	Chemical Engineering Department	Date:	10/22/2019		
Utility Class	ASPEN ID	Service	Tag	Usage	Unit Cost	Annual Cost, \$/yr
Cooling Water		Cooling of Heat Exchangers	CW-1	32.6714 KLB/H	\$295.95	\$2,323,295.52
Power		Powering Pumps and Compressor	P-1	52.32 kW	\$160.24	\$58,528.80
Cyclohexane Production		Selling Product	C-1	303.37 KMTA	\$1,044	\$316,718,280
Benzene Expense		Benzene Cost	B-1	288.03 KMTA	\$753	-\$216,886,590
H2 Expense		H2 Cost	H-1	22.68 KMTA	\$13	-\$9,216,960
					Total Revenue	\$90,614,730
					Methane Credit	\$2,857,633
					Net Revenue	\$93,472,363

Table 5 – Cyclohexane Production Operating Cost Estimate

References

Marshall and Swift Cost Index, M&S